**Unit:** Unit 8: Quadratics
**Topic:** Vertex Form and Intercept Form

**Name:** ________________
**Date:** ____________

# Foundation (Jalapeno

**Q1.** The vertex form of a quadratic function is $y = a(x-h)^2+k$. What does $(h, k)$ represent?
(A) The $x$- and $y$-intercepts
(B) The vertex of the parabola
(C) The axis of symmetry
(D) The domain and range
(E) The roots of the equation

**Q2.** Which of the following is the vertex form of the quadratic function $y = x^2 + 4x + 4$?
(A) $y = (x+2)^2$
(B) $y = (x-2)^2$
(C) $y = (x+1)^2 + 1$
(D) $y = (x-1)^2 - 1$
(E) $y = x^2 + 4x + 4$

**Q3.** The equation of a quadratic function in intercept form is $y = a(x-p)(x-q)$. What do $p$ and $q$ represent?
(A) The $x$-intercepts
(B) The $y$-intercepts
(C) The vertex of the parabola
(D) The axis of symmetry
(E) The domain and range

**Q4.** Which of the following is the intercept form of the quadratic function $y = x^2 - 5x - 6$?
(A) $y = (x+1)(x-6)$
(B) $y = (x-1)(x+6)$
(C) $y = (x+2)(x-3)$
(D) $y = (x-2)(x+3)$
(E) $y = x^2 - 5x - 6$

**Q5.** The quadratic function $y = a(x-h)^2 + k$ has its vertex at $(h, k) = (2, 3)$. What is the equation of the axis of symmetry?
(A) $x = -2$
(B) $x = 2$
(C) $x = 3$
(D) $y = 2$
(E) $y = 3$

**Q6.** The quadratic function $y = a(x-p)(x-q)$ has $x$-intercepts at $(p, 0) = (1, 0)$ and $(q, 0) = (-2, 0)$. What is the equation of the parabola?
(A) $y = (x-1)(x+2)$
(B) $y = (x+1)(x-2)$
(C) $y = x^2 + x - 2$
(D) $y = x^2 - x - 2$
(E) $y = x^2 - 3x + 2$

**Q7.** The quadratic function $y = ax^2 + bx + c$ can be written in vertex form as $y = a(x-h)^2 + k$. What is the relationship between $a$, $b$, and $c$ in the two forms?
(A) $h = -\frac{b}{2a}$ and $k = c - \frac{b^2}{4a}$
(B) $h = \frac{b}{2a}$ and $k = c + \frac{b^2}{4a}$
(C) $h = -\frac{b}{2a}$ and $k = c + \frac{b^2}{4a}$
(D) $h = \frac{b}{2a}$ and $k = c - \frac{b^2}{4a}$
(E) $h = \frac{b}{a}$ and $k = c - \frac{b^2}{a}$

**Q8.** The quadratic function $y = x^2 - 4x - 3$ can be written in intercept form as $y = a(x-p)(x-q)$. What are the values of $p$ and $q$?
(A) $p = -1$ and $q = 3$
(B) $p = 1$ and $q = -3$
(C) $p = -3$ and $q = 1$
(D) $p = 3$ and $q = -1$
(E) $p = 2$ and $q = -2$

**Open Ended Questions (FRQ)**

**Q9.** Write the quadratic function $y = x^2 + 2x - 6$ in vertex form $y = a(x-h)^2 + k$. Identify the values of $a$, $h$, and $k$.

**Q10.** Write the quadratic function $y = x^2 - 7x + 10$ in intercept form $y = a(x-p)(x-q)$. Identify the values of $p$ and $q$ and the $x$-intercepts.

**Chef's Tips**

- Tip 1: Make sure to simplify all expressions and equations.
- Tip 2: Check for any common factors before factoring.
- Tip 3: Use the vertex form and intercept form to analyze the graph of a quadratic function.